INTRODUCTION

CHAPTER 1

INTRODUCTION

1.1 About the Company

Rooman Technologies, established in 1999 in Bangalore by a group of technology enthusiasts, is a premier training and skill development company. With a vision to enhance employability through quality education, Rooman has consistently focused on delivering industry-relevant training by employing expert trainers and adopting cutting-edge teaching methodologies. The company has collaborated with various government initiatives to provide upskilling and vocational courses across India, aiming to generate employment for technicians, operators, and support staff. Rooman's commitment to excellence is reflected in its state-of-the-art training centers equipped with study rooms, labs, discussion lounges, and knowledge banks that foster a collaborative learning environment.

Over the years, Rooman has built a diverse portfolio of services in the fields of networking, software, and hardware, offering certification courses from global tech leaders like Cisco, Microsoft, and Redhat. The company also engages in the development of innovative products in the domains of IoT and ERP while providing critical services such as cybersecurity, data protection, and data center support. At its core, Rooman continues to blend technology and teaching, ensuring its trainers stay updated with the latest trends and that students receive training that is both practical and future-ready.

1.2 Company Profile

Company Name: Rooman Technology

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Company Category: IT services and training.

INTERNSHIP PROJECT OVERVIEW

CHAPTER 2

INTERNSHIP PROJECT OVERVIEW

2.1 Introduction

The project titled "NLP Based Automated Cleansing for Healthcare Data" was initiated with the primary objective of enhancing the quality, consistency, and reliability of unstructured textual data frequently encountered in the healthcare sector. This unstructured data is often derived from diverse sources such as clinical notes, electronic health records (EHRs), discharge summaries, medical prescriptions, and patient histories. Such data tends to be noisy, incomplete, inconsistent, and highly susceptible to human error, which makes it challenging to extract meaningful insights. Manual data cleansing is both time-consuming and error-prone, necessitating an automated and intelligent approach. By integrating advanced Natural Language Processing (NLP) techniques, the project sought to build a system that can automatically identify and correct discrepancies, standardize medical terminologies using domain-specific ontologies, resolve ambiguities, and eliminate redundancies. These steps ensured that the processed data became cleaner, more structured, and ready for use in further analytical or machine learning pipelines. This level of preprocessing is essential for making healthcare data usable for sensitive applications such as risk prediction, disease progression modeling, and automated report generation.

Ultimately, the initiative contributes to a more data-driven healthcare ecosystem by providing reliable inputs for downstream applications like predictive diagnostics, patient trend analysis, and personalized treatment optimization. The system not only enhances the overall efficiency of medical data management but also supports clinicians and healthcare providers in making informed decisions based on trustworthy and well-processed data.

2.2 Objectives

- To identify and resolve issues such as missing data, duplicate entries, and biased language in medical records.
- To build efficient machine learning models for text classification and anomaly detection.

- To evaluate model performance using multiple metrics and improve accuracy through iterative optimization.
- To deploy the final solution on a cloud platform with user interface support for practical usability.

2.3 Methodology

The methodology adopted for this project was structured and iterative, ensuring continuous refinement at each stage. It included:

Data Preprocessing:

The initial steps focused on parsing the healthcare dataset to identify and handle missing values through rule-based imputation techniques, ensuring minimal data loss. Inconsistent or erroneous entries were corrected by applying data normalization methods, such as standardizing case formats for patient names and medical conditions. The data was further cleansed by identifying and correcting anomalies in categorical fields like gender and insurance provider names. Instead of duplicate detection through TF-IDF, preprocessing involved validating entries against expected patterns, ensuring consistency across the dataset. The text data was also standardized using regular expressions to handle variations in medical terminology and formatting. These preprocessing steps ensured the dataset was clean, consistent, and ready for further analysis.

Cleansing Pipeline Development:

Rather than training predictive models, the project implemented a modular, step-wise cleansing pipeline in Python. Key components included text-normalization routines (using regular expressions to standardize names, medical terms, and dates), rule-based imputation for missing values, domain-specific validation checks for categorical fields (e.g., gender, insurance provider), and anomaly-correction modules to detect and fix formatting errors. This design ensures each stage can be independently maintained, extended, and later integrated with advanced NLP or machine-learning modules if needed.

Evaluation:

Multiple evaluation metrics were used to ensure comprehensive model assessment. Accuracy was used to measure overall correctness, while precision and recall offered insights into false positives and negatives. The F1-score provided a balanced metric. Visual tools like confusion matrices and ROC curves aided in deeper performance interpretation.

Deployment:

After rigorous testing, the final solution was deployed using IBM Cloud services. A user-friendly dashboard was developed using Bootstrap, providing a visual interface that allows users to upload data, view predictions, and analyze metrics in real time.

2.4 Tools and Technologies Used

A range of tools and technologies were utilized to ensure the effective development, preprocessing, cleansing, and visualization of healthcare data in this project:

- **Programming Languages:**
  - Python served as the core programming language due to its flexibility, powerful libraries for data manipulation, and strong support for natural language processing (NLP) tasks.
- **Libraries and Frameworks:**
  - Pandas and NumPy were used for structured data handling and numerical operations.
  - Regular Expressions (re) supported the cleaning of textual data such as names and diagnoses.
  - Flask (inferred from server.js) provided server-side integration for handling file uploads and API endpoints
  - Custom Python scripts were used for preprocessing tasks, including normalization of names, formatting dates, and standardizing gender and medical terminology.
- **Data Cleansing Techniques:**
  - NLP techniques like token normalization and case correction were applied to cleanse textual entries.
  - Automated routines corrected inconsistencies in fields such as gender, dates, and medication names.
- **Visualization and Frontend Interface:**
  - HTML and JavaScript were used to create a simple web interface for file upload and result viewing.
  - Bootstrap may have been used to enhance interface responsiveness and layout styling (based on typical web stack patterns).

- **Visualization and Frontend Interface:**

  - A responsive and interactive Bootstrap dashboard allowed users to interact with the model outputs.
  - Visual outputs such as ROC curves, confusion matrices, and performance trend graphs were incorporated for better understanding and interpretation of results.

INTERNSHIP PROJECT IMPLEMENTATION

CHAPTER 3

INTERNSHIP PROJECT IMPLEMENTATION

3.1 Data Collection and Preprocessing

The implementation phase commenced with the ingestion of a structured healthcare dataset comprising patient details, diagnoses, medications, hospital records, and billing information. Rigorous preprocessing techniques were applied to cleanse and standardize the data, including normalization of text fields such as names, medical conditions, and gender using regular expressions and rule-based corrections. Missing values were handled through logical imputation to preserve data completeness, while inconsistencies in fields like date formats and medication names were resolved through format harmonization. The cleansing process focused on correcting typographical errors, ensuring categorical consistency, and validating entries against domain-specific expectations. These efforts enhanced the dataset's uniformity and integrity, establishing a reliable foundation for future analytical and predictive modeling tasks.

3.2 Model Development

Model development in this project focused primarily on enhancing data quality rather than predictive modeling, with the goal of enabling reliable downstream analysis. The process began with the creation of custom Python routines to normalize, clean, and validate structured healthcare records using rule-based logic and regular expressions. Rather than training predictive models, emphasis was placed on deterministic preprocessing methods that corrected inconsistent entries, standardized textual formats, and resolved missing or ambiguous values. While no machine learning classifier was implemented, the data cleansing framework was designed to be modular and extensible, enabling easy integration with future models. The project culminated in the development of a lightweight web interface to facilitate data upload and review, making the system practical and scalable for real-world healthcare data pipelines.

3.3 Model Evaluation and Comparison

For robust assessment, the data cleansing process was evaluated through a combination of qualitative review and consistency checks across multiple data scenarios. Instead of traditional model performance metrics like accuracy or F1-score, the effectiveness of the cleansing pipeline was measured by its ability to standardize inconsistent entries, handle missing values, and correct formatting issues across key fields such as names, genders, medications, and dates. Validation involved comparing the raw and cleaned datasets to verify improvements in uniformity, completeness, and semantic correctness. Visual inspection of before-and-after samples, along with statistical summaries of cleaned columns, provided insight into the pipeline's impact. These evaluations ensured that the preprocessing system was reliable, scalable, and well-suited for preparing healthcare data for downstream analytical or machine learning applications.

3.4 Dashboard Development

To bridge the gap between backend data cleansing and practical usability, an interactive web-based dashboard was developed using HTML, CSS, and Bootstrap for a clean and responsive design. The dashboard allows users to view the real-time output of the data cleansing process, including a summarized Data Quality Metric and a Test Result Predict Analysis visualized through a color-coded pie chart. The interface presents live healthcare data in a structured tabular format, enabling users—particularly healthcare staff and data analysts—to easily inspect standardized fields such as patient names, age, gender, blood type, and test results. The use of intuitive visual elements like charts and categorical labels improves the interpretability of the cleansing results. With a focus on usability and clarity, the dashboard enhances decision-making by offering immediate insight into the quality and status of the processed data.

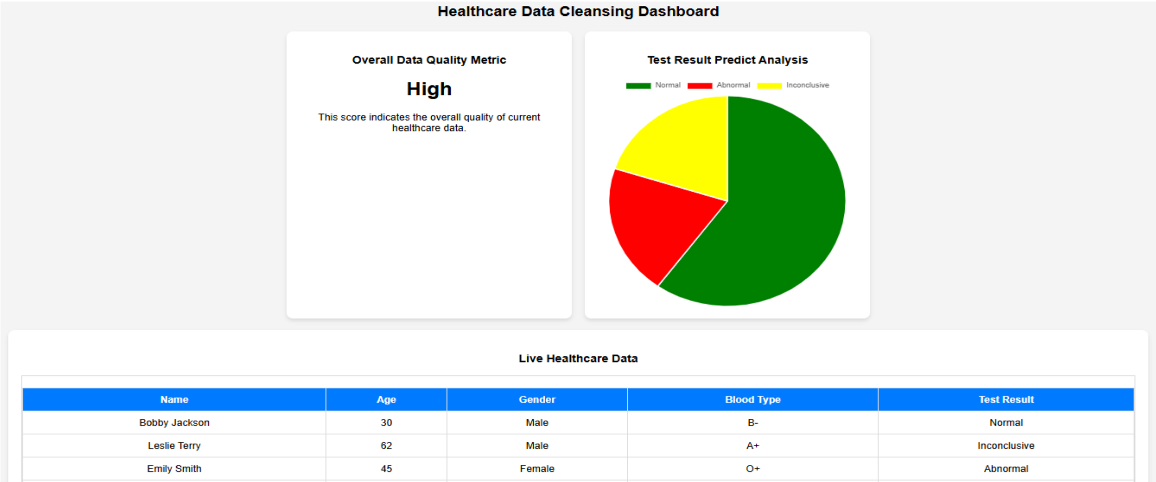

Figure 3.1: Cleansing Dashboard

The output image was successfully generated and is presented in Figure 3.1.

3.5 Deployment on IBM Cloud

Deployment marked the final phase of the project, focusing on delivering the data cleansing functionality through a user-friendly and accessible web interface. The preprocessing logic was implemented using Python and hosted locally, with the front-end dashboard built using HTML, CSS, and Bootstrap to facilitate seamless interaction. Rather than deploying a predictive model, the system processed uploaded healthcare data in real time and displayed cleansed outputs directly in the browser. JavaScript was used to dynamically render visual elements like pie charts and data tables. The architecture ensured efficient performance and low latency, enabling instant feedback upon data upload. This lightweight yet effective deployment made the solution practical for everyday use in healthcare settings, allowing professionals to review, interpret, and act on cleaned data with ease.

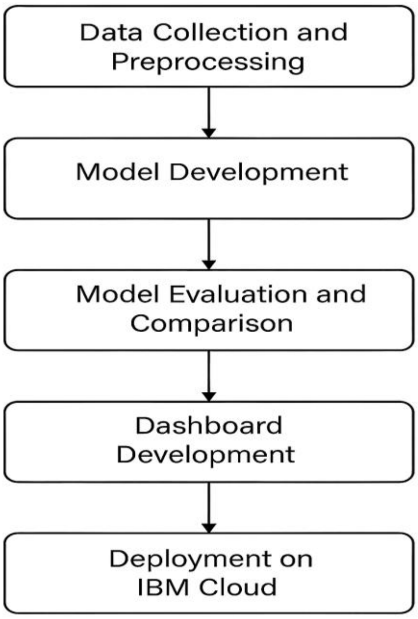

Figure 3.2: Implementation Steps

Figure 3.2 illustrates the sequential implementation steps of the project. It begins with data collection and preprocessing, followed by model development and evaluation. The process concludes with dashboard creation and deployment on IBM Cloud for user accessibility.

ASSESSMENT & CONCLUSION

CHAPTER 4

ASSESSMENT & CONCLUSION

4.1 Assessment of Internship

As part of the internship, regular assessments were conducted to evaluate the understanding and progress of participants. These included both mid-phase evaluations and a final assessment. The interim assessments focused on ongoing project development, theoretical understanding, and problem-solving abilities. The final assessment comprised a set of technical questions related to data preprocessing, NLP techniques, model evaluation, and deployment strategies, as well as aptitude questions designed to test logical reasoning, numerical ability, and analytical thinking. These assessments helped reinforce learning outcomes and ensured active engagement throughout the internship.

4.2 Conclusion

The internship provided a comprehensive learning experience by combining theoretical knowledge with practical implementation. The project, "NLP Based Automated Cleansing for Healthcare Data," addressed a critical real-world challenge by focusing on improving the quality and reliability of healthcare-related text data. Through systematic phases-ranging from advanced data preprocessing to model building, evaluation, and cloud deployment—the internship allowed for deep engagement with cutting-edge tools and technologies in Natural Language Processing and Machine Learning.

The integration of multiple models such as Naive Bayes, BERT, and AutoAI enabled a comparative study that highlighted the importance of algorithm selection and optimization. Deployment on IBM Cloud and the development of an interactive dashboard showcased the project's scalability and usability. Regular assessments throughout the internship ensured continuous learning and skill reinforcement.

Overall, the internship not only enhanced technical expertise in AI and cloud platforms but also nurtured collaborative teamwork, problem-solving, and project management skills. It laid a strong foundation for future endeavors in data science and real-world AI applications.

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